RE-303 CPU

SONIC-POTIONS

!!! PRELIMINARY DRAFT VERSION !!!

WORK IN PROGRESS

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Intro

The RE-303 CPU was developed for the DinSync.info RE-303 replica.

The basic sequencer concept is the same as on the original 303. Pitch and Time information is stored independently. There are time steps that only hold **step**(note on) **tie** (note length) and **rest** (no note) information. The pitch steps store the note value and accent, slide as well as transpose information.

Quickstart

Control scheme

The 303 has four modes: Write Track, Play Track, Write Pattern, Play Pattern.

Each mode has multiple control layers that give the buttons and LEDs other functions. These are **normal mode**, **pitchmode** and **time mode.** These modes are accessed by pressing or holding the corresponding button.

Normal Mode

Pitch Mode

Time Mode

Pattern Write Mode

In this mode you create patterns.

The classic way

- Set pattern length
- enter pitch
- enter time

Pattern length

There are various ways to set the pattern length (1 to 16 Steps). Length is edited in **function mode**. This mode is active while the **FUNCTION** button is held down.



Here are the track length key combos:

Hold FUNCTION = show pattern length

FUNCTION + DOWN = tap pattern length (8xDOWN => pattern length 8)

FUNCTION + **note buttons** = set pattern length directly. white = 1-8/8-16 depending on C# or D#

FUNCTION + BACK/NEXT = decrease/increase pattern length one step at a time

FUNCTION + ACCENT = half pattern length

FUNCTION + SLIDE = double pattern length

Triplets

Triplet mode is used to create patterns in a 3/4 measure. In the normal 4/4 measure a beat is divided into

4/4 = 16 steps = 6 clock ticks per step

¾ = 12 steps = 8 clock ticks / step

To turn triplet mode on or off press FUNCTION + UP

Pattern Chain

hold down first white note/pattern button to select chain start

then press 2nd white note/pattern button to select chain end.

All patterns in between will be played after one another

Chained pattern LEDs will light up, playing pattern will blink

In pattern Write mode this also works for live recording mode!



i.e you can chain 4 patterns to record a 4 bar pattern in one go.

Normal Mode

/* - Normal Mode LED lit

* - white notes show selected pattern blinking

* - active chain/next pattern lit

- chaselight

* - Up/Down show direction

- A/B pattern section

*/

Pattern Direction

The pattern playback direction can be changed and is saved with the pattern.

in normal mode:

Down = back

Up = forward

Up+Down = RND

Down + Up = PingPong

Time Mode

- time mode LED lit

- keys show step number (white keys 1-8, black keys sets of 8)

- down, up, accent = step, tie, rest

Active steps, ties and rests are placed in a normal 16 step sequencer.

Pitch mode

/* - Pitch mode LED lit

* - steps indicate selected pitch of active steps

* - accent, slide, up/down show active attributes

*/

Pitch, accent, slide and transpose are set in the pitch sequencer.

When the seq is stopped, entered notes are played.

Note entry

- Step buttons enter notes

- auto advance

- hold acc,slide,up/down while pressing note to add attributes

Note attributes

- hold next or back to show step information and step through the sequence

- pitch is shown on step buttons

- attributes are shown

- attributes can be toggled with their buttons

Live Recording Mode

- no mode led lit

- chaselight

Allows to record notes into the running sequencer either by playing the 303s note buttons or a MIDI controller.

To enter live recording mode press the **BACK** button while the sequencer is running in normal mode. To return to normal mode pres **BACK** or the **FUNCTION** button.

Recording with internal buttons

- record notes using note buttons

- add accented or transposed notes by holding acc/up/down before pressing note button

- slides are recorded by playing legato

Recording MIDI

MIDI can be used to input notes. A note on velocity above 65 plays an accented note. To add slides play 2 notes legato.

Classic 303 Tap Time Mode

The **TAP** button can be used to tap in time information (step and tie events)

live pitch mode

- Pitch led lit

- chaselight

- accent slide up down

-

- hold pitch -> live pitch mode

- acc/slide/up/down can be toggled and viewed via the buttons

Chain Recording

If a pattern chain is activated before entering live recording mode, sequences up to 8 patterns (128 Steps) can be recorded in realtime.

Clear Mode

Clear mode is active while the **CLEAR** button is held.

Pattern Randomizer

A random pattern can be generated in the current active slot.



CLEAR + ACCENT = randomize all

CLEAR + TIME + ACCENT = randomize time

CLEAR + PITCH + ACCENT = randomize pitch

Clear Pattern

To clear pitch and time information on the selected pattern, press **CLEAR + PATTERN**

Press **CLEAR + TIME + PATTERN** to only delete the time information.

Press **CLEAR + PITCH + PATTERN** to only delete the time information.

Pattern Copy/Paste

A Pattern can be copied to another slot.

CLEAR + C# + PATTERN copies the pattern data from the selected pattern slot.

CLEAR + D# + PATTERN pastes the previously copied data to the selected pattern slot.

CLEAR + D# + Pitch + PATTERN = paste pitch info

CLEAR + D# + Time + PATTERN = paste time info

Pattern Play Mode

Normal Mode

/* - Normal Mode LED lit

* - white notes show playing pattern (blinking)

* - light next pattern/chain

- pattern section

*/



Pitch Mode

- led pitch

- steps = transpose

- acc, slide up down

AllAccent + AllSlide

Pitch + acc = allAccent

Pitch + Slide = All Slide

Transpose pattern up/down

Pitch + up = transpose octave down

Pitch + down = transpose oct up

Pitch + Note buttons = transpose semitones

Pattern Chain

hold down first white note/pattern button to select chain start

then press 2nd white note/pattern button to select chain end.

All patterns in between will be played after one another

Chained pattern LEDs will light up, playing pattern will blink

Track Write Mode

Normal Mode

- next/back select track step to edit

- note buttons select assigned pattern

- acc/slide select pattern section A/B

- pattern chains can be assigned by holding 1st pattern button and pressing a 2nd one

Pitch Mode

Hold pitch -> pitch mode

- note buttons = semitone transpose (track step transpose, independent of pattern transpose value)

- up/down => octave transpose

Time Mode

Hold Time => time mode

- leds show track step repeats (1-16)

- last step indicator => ACC,SLIDE;UP,DOWN lit

- next/back inc/dec repeats

- down = 1 repeat, up=2, acc=4, slide=8

- time + clear = set current step as last step

- note buttons select repeats (white notes 1-8, black notes = range 1-8 or 9-16)

Clear Mode

Clear Track

- clear + next, then back = clear track

Copying tracks

- C# -> track copy

- D# -> track paste

Track Play Mode

############## Track Play ####################

- leds show playing/active pattern

- hold tap/next => show playing track step (realtime update)

- tap+back -> restart track

hold pitch -> pitch mode

- all accent, all slide

- LEDs show current transpose value (realtime update)

hold time -> time mode

- leds show repeats (realtime update)

- leds show playing/active pattern

- hold tap/next => show playing track step (realtime update)

- tap+back -> restart track

Pitch Mode

Pitch mode is active while the **PITCH** button is held down.

- all accent, all slide

- LEDs show current transpose value (realtime update)

Time Mode

hold time -> time mode

- leds show repeats (realtime update)

MIDI control

While the sequencer is stopped the 303 can be played via MIDI note messages.

Accent

If the MIDI note on velocity >= 65 the note is played accented.

Slide

Legato play of 2 notes causes a slide

Config Mode

FUNCTION + CLEAR => config mode



The config is saved when leaving config page (press function button again to leave)

MIDI Channel

DOWN = MIDI Rx channel can be selected with next/back keys. channel displayed on LEDs

UP = MIDI channel can be selected with next/back keys. channel displayed on LEDs

MIDI Rx

C = MIDI RX on/off

MIDI Sync Rx

D = MIDI sync rx on/off

MIDI Tx

E = MIDI Tx On/off

Timing

F = Internal timing

on = Original Mode

delayed gate offs

off = tight mode

Gate on/off completely same length (3 clocks on, 3 off)

Saving data

Pattern and track data is auto saved whenever the sequencer is stopped and automatically loaded on power on.

Firmware Update

The firmware can be updated via MIDI SysEx. To update the firmware:

- hold CLEAR while powering up
- the first 4 step LEDs will light up
- The unit is now ready to receive firmware via sysex
- While the data is transmitted the progress is shown on the step LEDs.
- Once the update is done the machine will reboot and the new firmware is ready to use

MIDI implementation chart

Manufacturer: Sonic Potions	Model: Re-303 CPU	Version: 1.0	Date: 01.16.2017	
		Transmit/Export	Recognize/Import	Remarks
1. Basic information				
MIDI channels		1-16	1-16	
Note numbers			0-63	Tx range?
Program change		N	N	
Bank select response? (Yes/No) If yes, list banks utilized in remarks column		N	N	
Modes supported:	Mode 1: Omni-On, Poly (Yes/No) Mode 2: Omni-On, Mono (Yes/No) Mode 3: Omni-Off, Poly (Yes/No) Mode 4: Omni-Off, Mono (Yes/No)			

	Multi Mode (Yes/No)			
Note-On Velocity (Yes/No)		Y	Y	Only used for accent.
Note-Off Velocity (Yes/No)				
Channel Aftertouch (Yes/No)				
Poly (Key) Aftertouch (Yes/No)				
Pitch Bend (Yes/No)				
Active Sensing (Yes/No)				
System Reset (Yes/No)				
Tune Request (Yes/No)				
Universal System Exclusive:	Sample Dump Standard (Yes/No)			
	Device Inquiry (Yes/No)			
	File Dump (Yes/No)			
	MIDI Tuning (Yes/No)			
	Master Volume (Yes/No)			
	Master Balance (Yes/No)			
	Notation Information (Yes/No)			
	Turn GM1 System On (Yes/No)			
	Turn GM2 System On (Yes/No)			
	Turn GM System Off (Yes/No)			
	DLS-1 (Yes/No)			
	File Reference (Yes/No)			
	Controller			

Destination (Yes/No) Key-based Instrument Ctrl (Yes/No) Master Fine/Coarse Tune (Yes/No) Other Universal System Exclusive		
Manufacturer or Non-Commercial System Exclusive	Y	Y
NRPNs (Yes/No)	N	N
RPN 00 (Pitch Bend Sensitivity) (Yes/No) RPN 01 (Channel Fine Tune) (Yes/No) RPN 02 (Channel Coarse Tune) (Yes/No) RPN 03 (Tuning Program Select) (Yes/No) RPN 04 (Tuning Bank Select) (Yes/No) RPN 05 (Modulation Depth Range) (Yes/No)		
2. MIDI Timing and Synchronization		
MIDI Clock (Yes/No)	Y	Y
Song Position Pointer (Yes/No)	Y	Y
Song Select (Yes/No)		
Start (Yes/No)	Y	Y
Continue (Yes/No)	Y	Y
Stop (Yes/No)	Y	Y
MIDI Time Code (Yes/No)		
MIDI Machine Control (Yes/No)		
MIDI Show Control (Yes/No) If yes, MSC Level supporte		
3. Extensions Compatibility		



General MIDI compatible? (Level(s)/No)
Is GM default power-up mode? (Level/No)
DLS compatible? (Levels(s)/No)
(DLS File Type(s)/No)
Standard MIDI Files (Type(s)/No)
XMF Files (Type(s)/No)
SP-MIDI compatible? (Yes/No)

